

FACULTY OF ECONOMIC SCIENCES, UW

Applications of **eXplainable AI** in Predictive Modelling

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XAI
2026

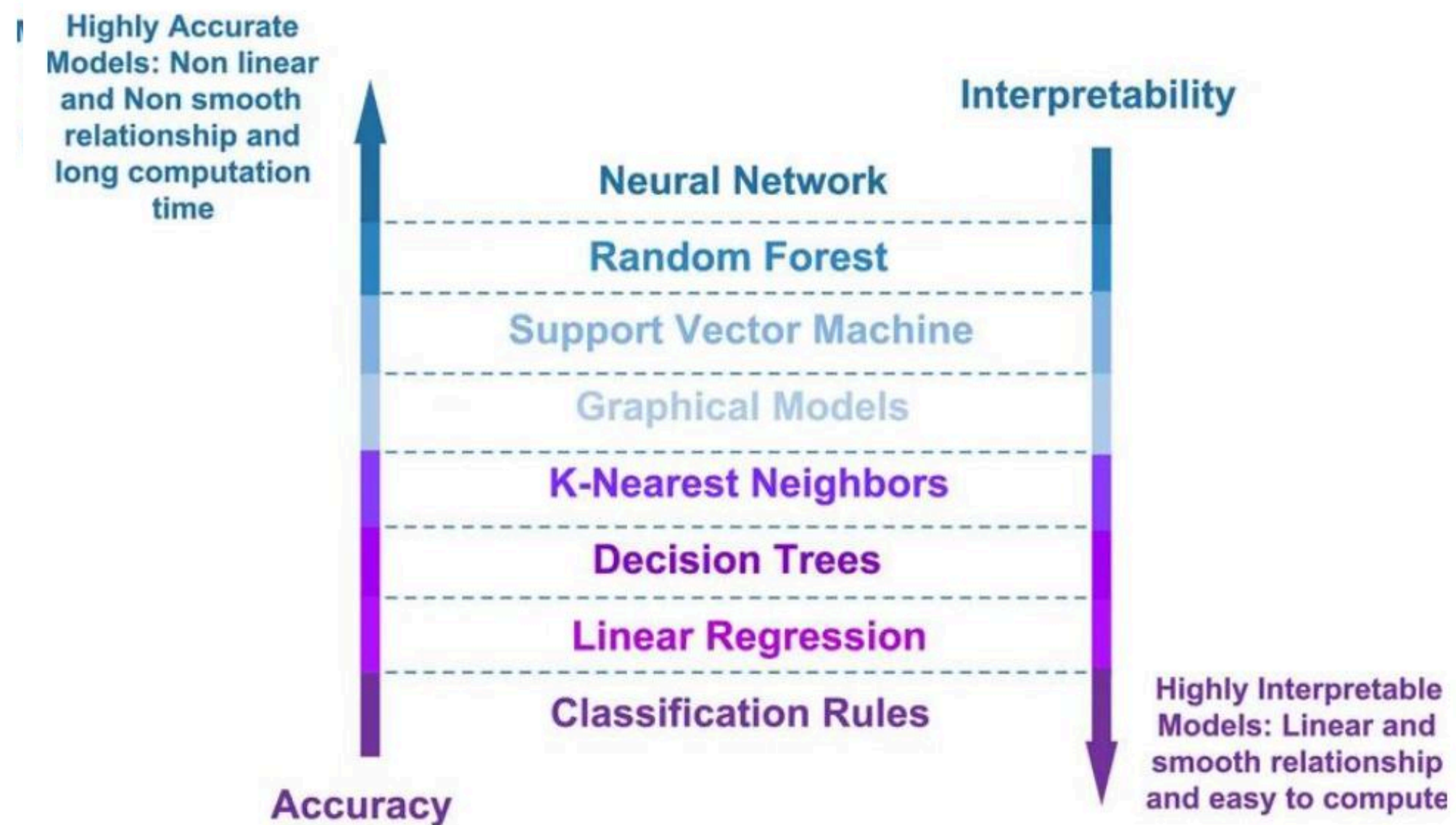
CH 01

Quick recap

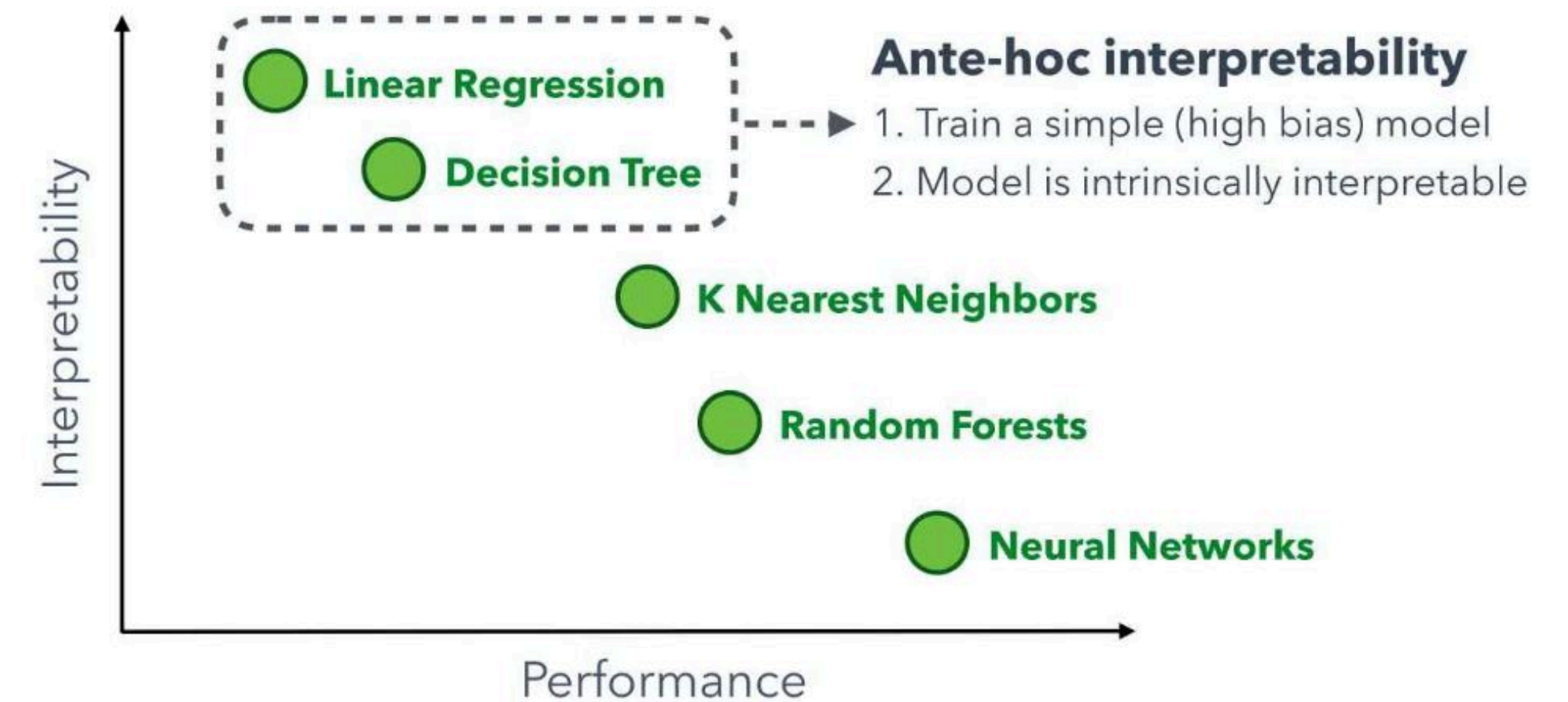
The taxonomy about XAI



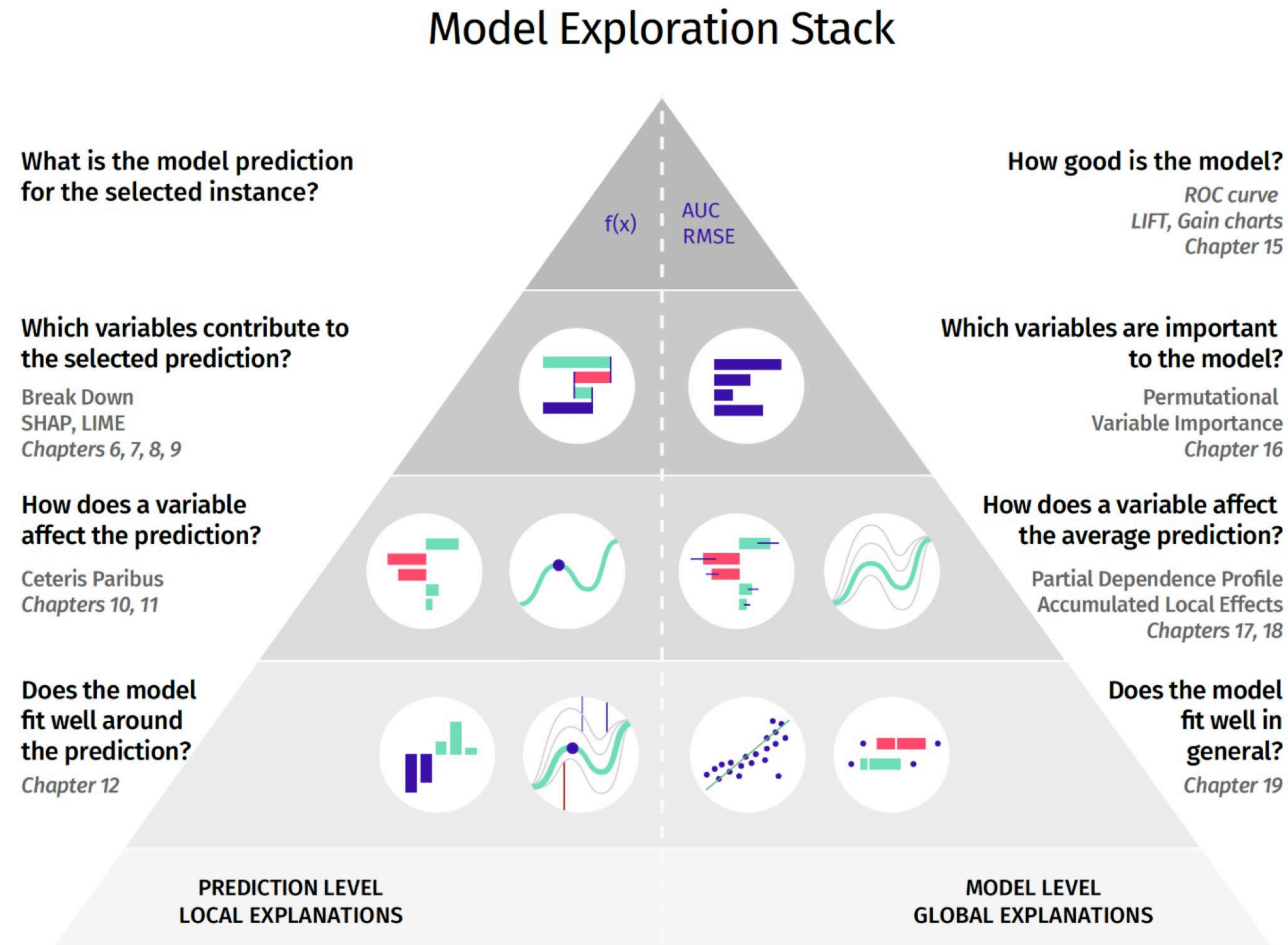
Model Interpretability vs Performance



Interpretability methods produce explanations



Model Interpretability vs Performance



Questions to answer with XAI

Correctness: Are we sure that all, and only, the features of interest contributed to our algorithm's decisions?

Robustness: Are we sure the model is not susceptible to disturbances?

Bias: Are we aware of any specific biases in the data that unfairly penalise groups of individuals?

Improvement: In what specific way can the prediction model be improved?

Transferability: Specifically how can the prediction model from one application domain be applied to another application domain?

Human understanding: Can we explain the model's algorithmic machinery to an expert or even to a layperson?



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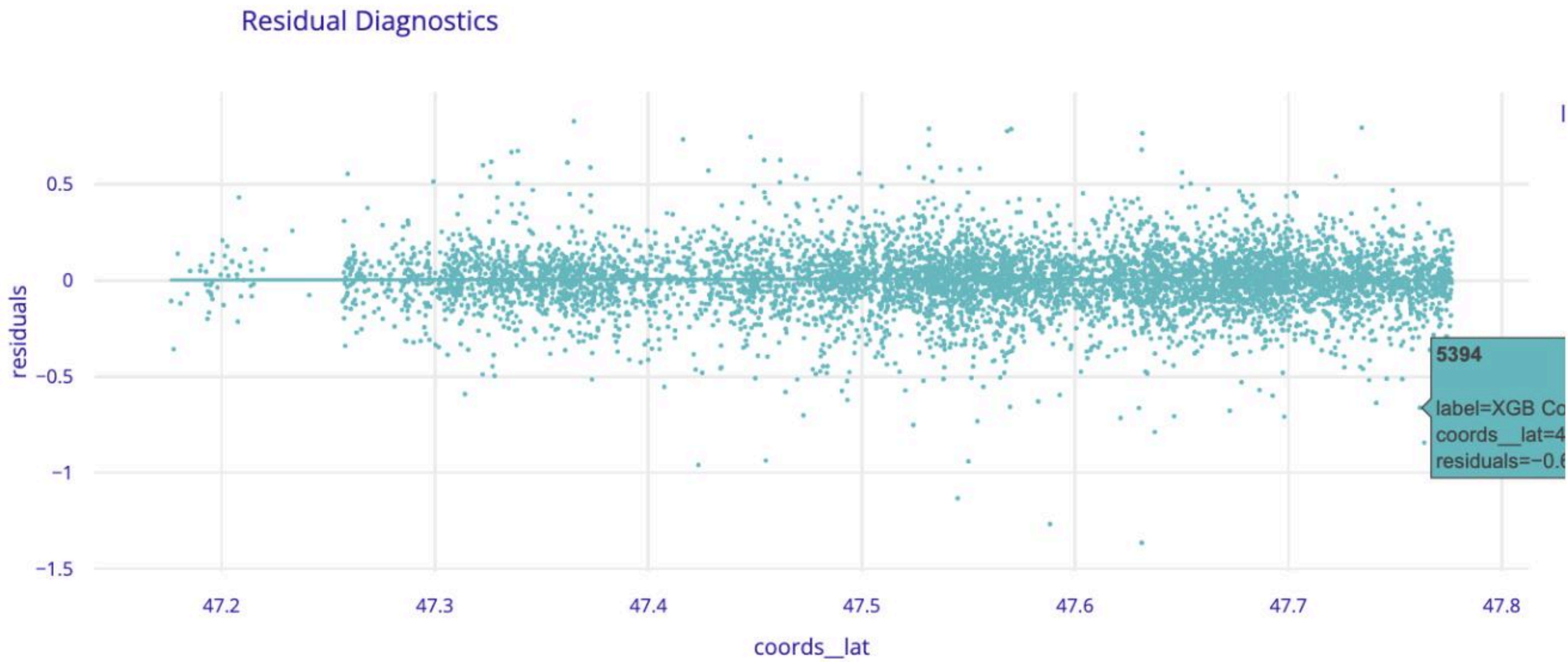
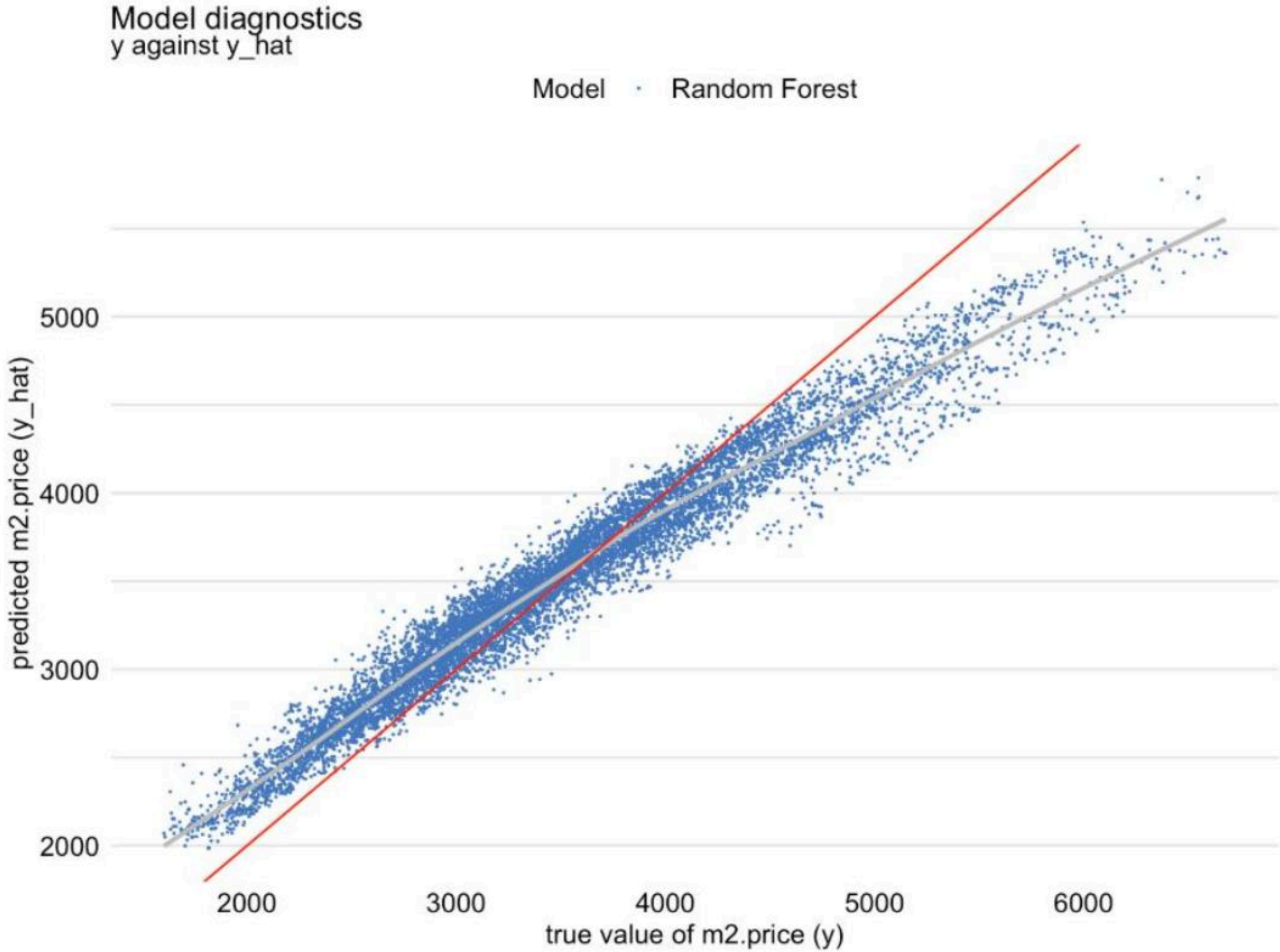
CH 02

Explainers for tabular data

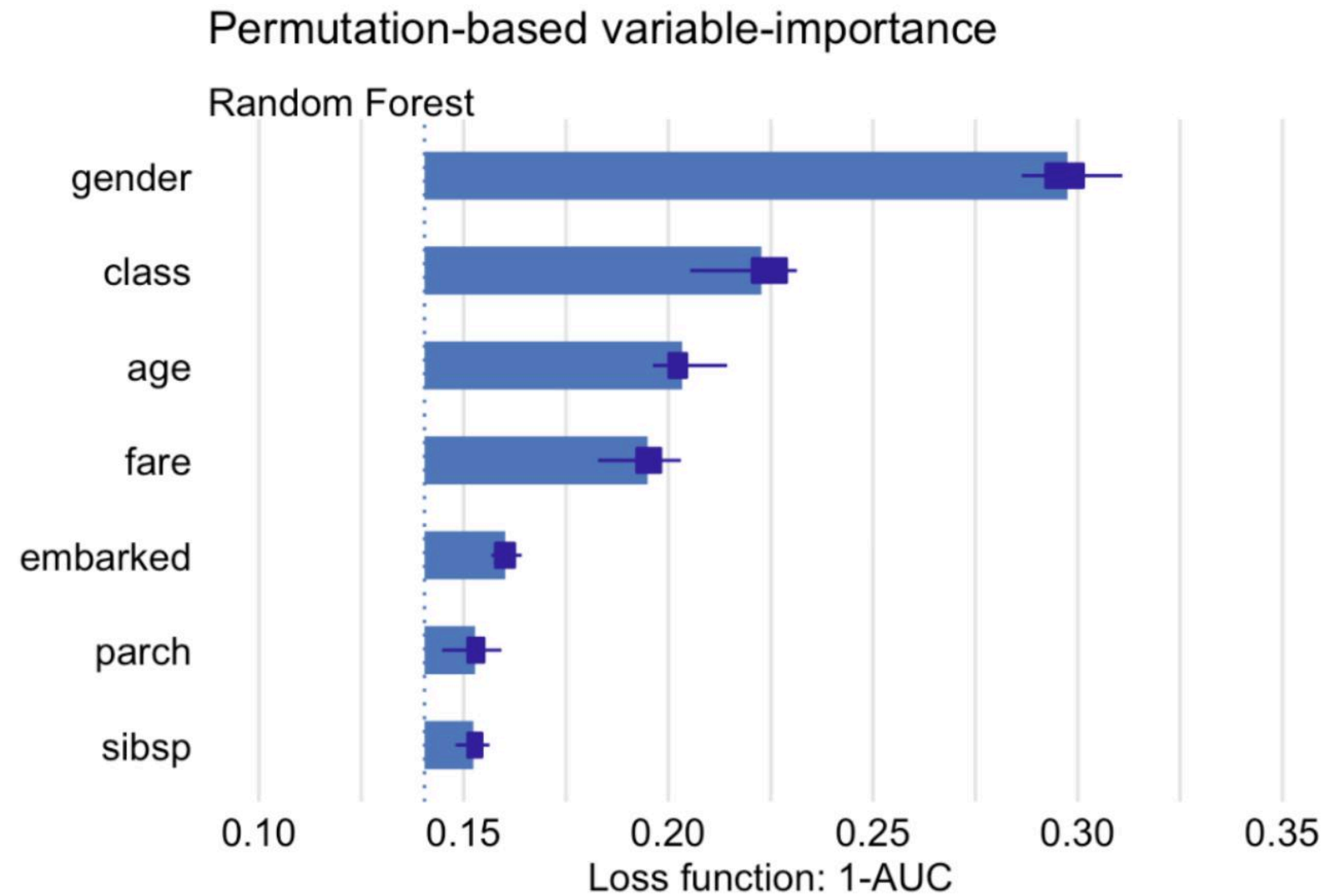
Core methods to remember



Residual Diagnostics

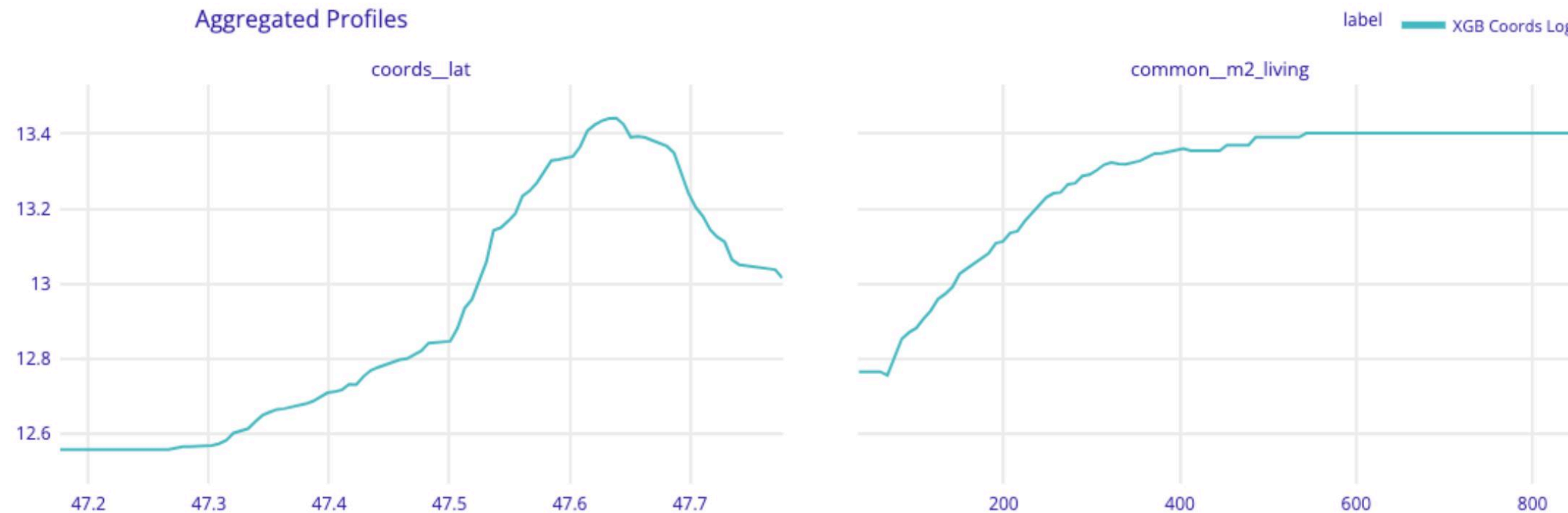


Permutation based feature importance

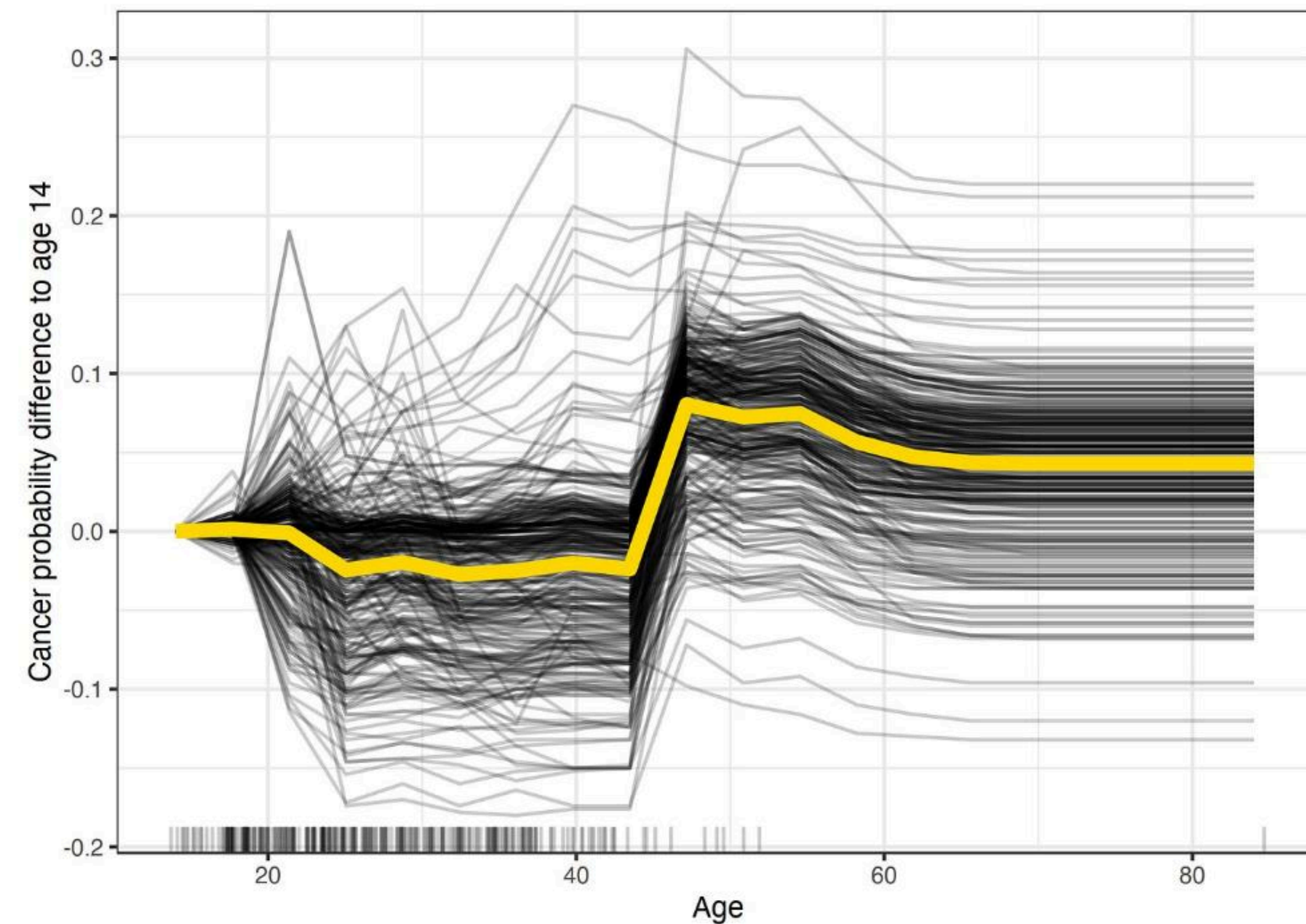


Partial dependence plots

The PDP (Partial Dependence Plot) shows the marginal effect that one or two features have on the predicted result of a machine learning model.

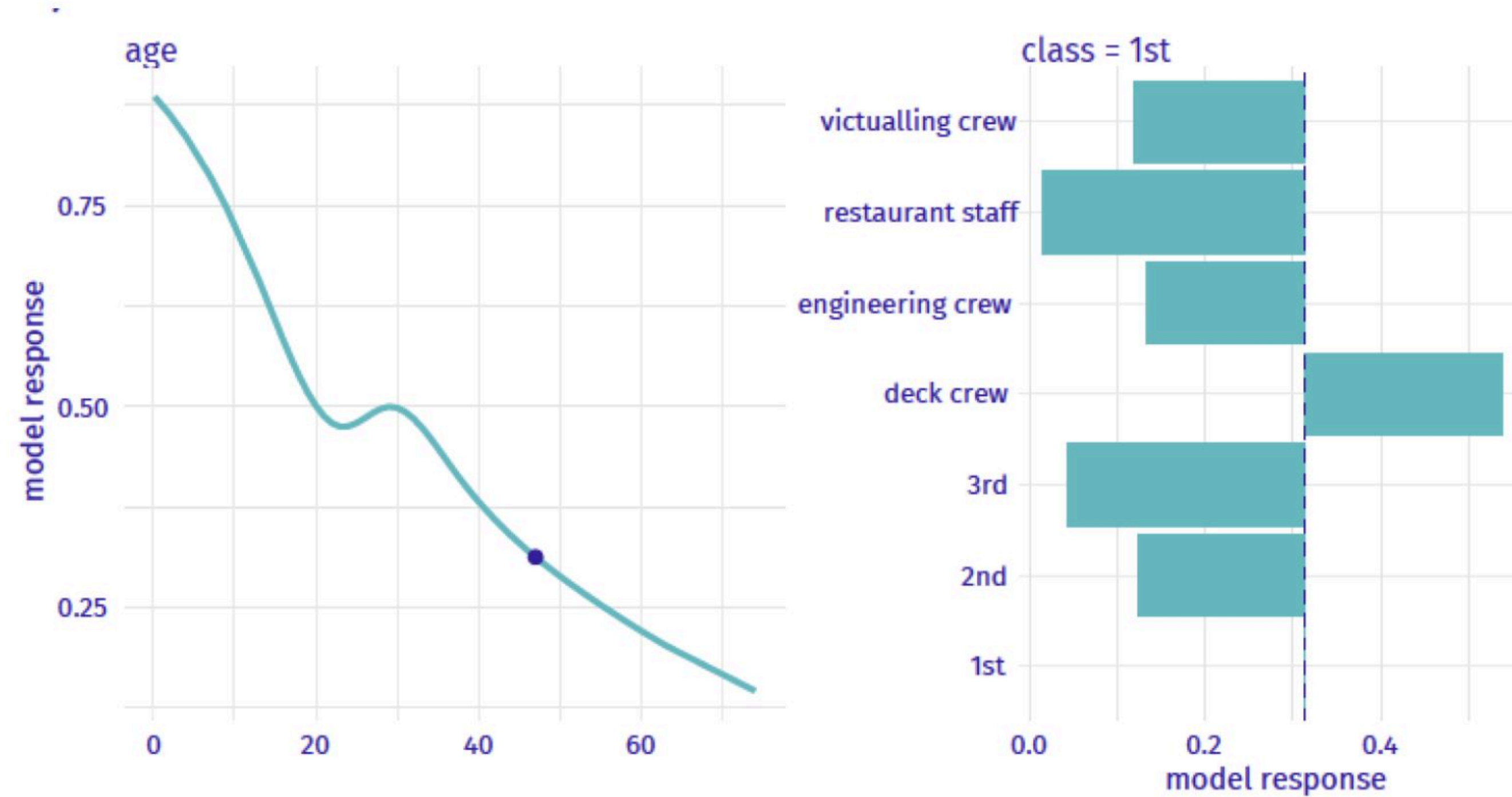


Individual conditional expectations (ICE)

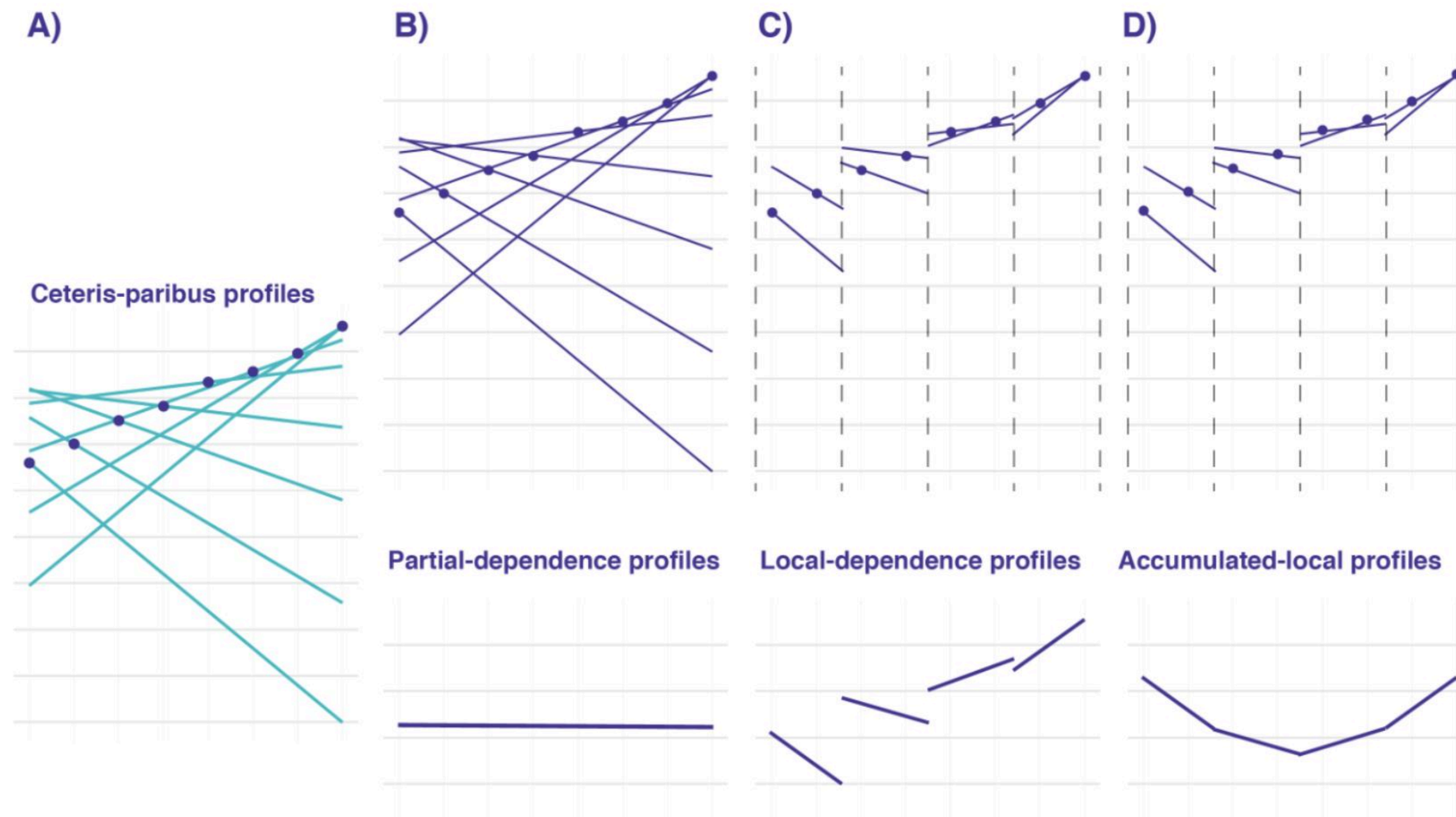


ICE (Individual Conditional Expectation) graphs show how the prediction of the instance changes when a feature changes. The Partial Dependence plot for the average effect of a feature is a global method because it does not focus on specific cases, but instead on a global average. The equivalent of a PDP for individual data instances is called an Individual Conditional Expectation (ICE) plot.

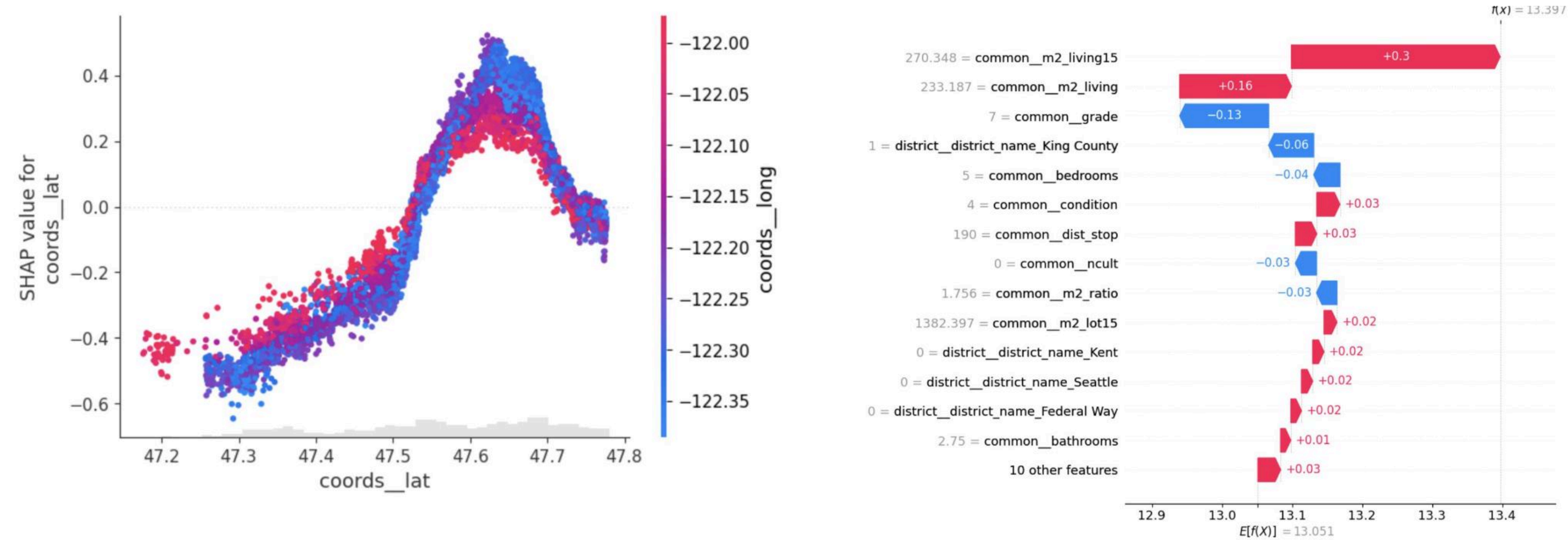
Ceteris paribus profiles



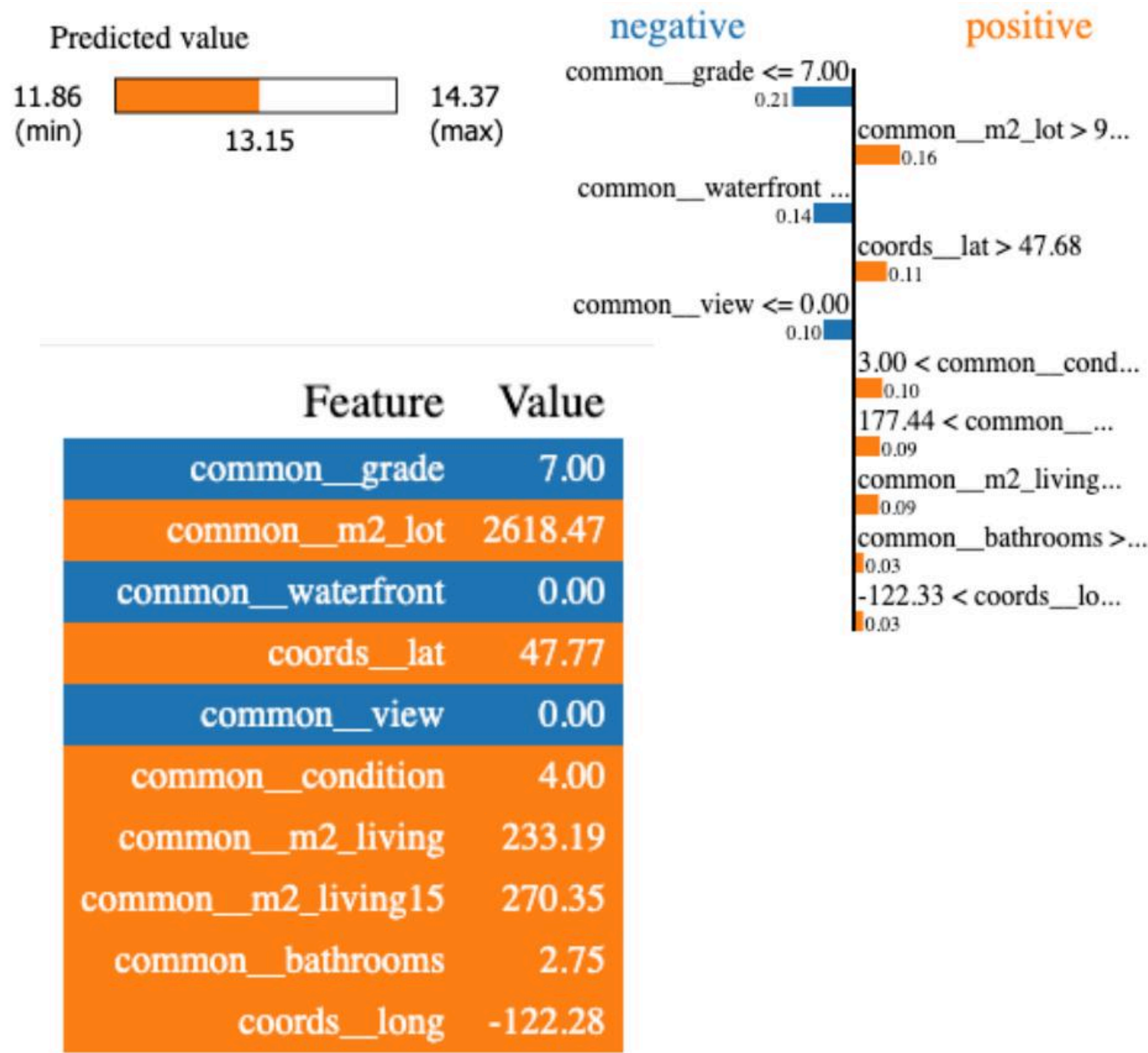
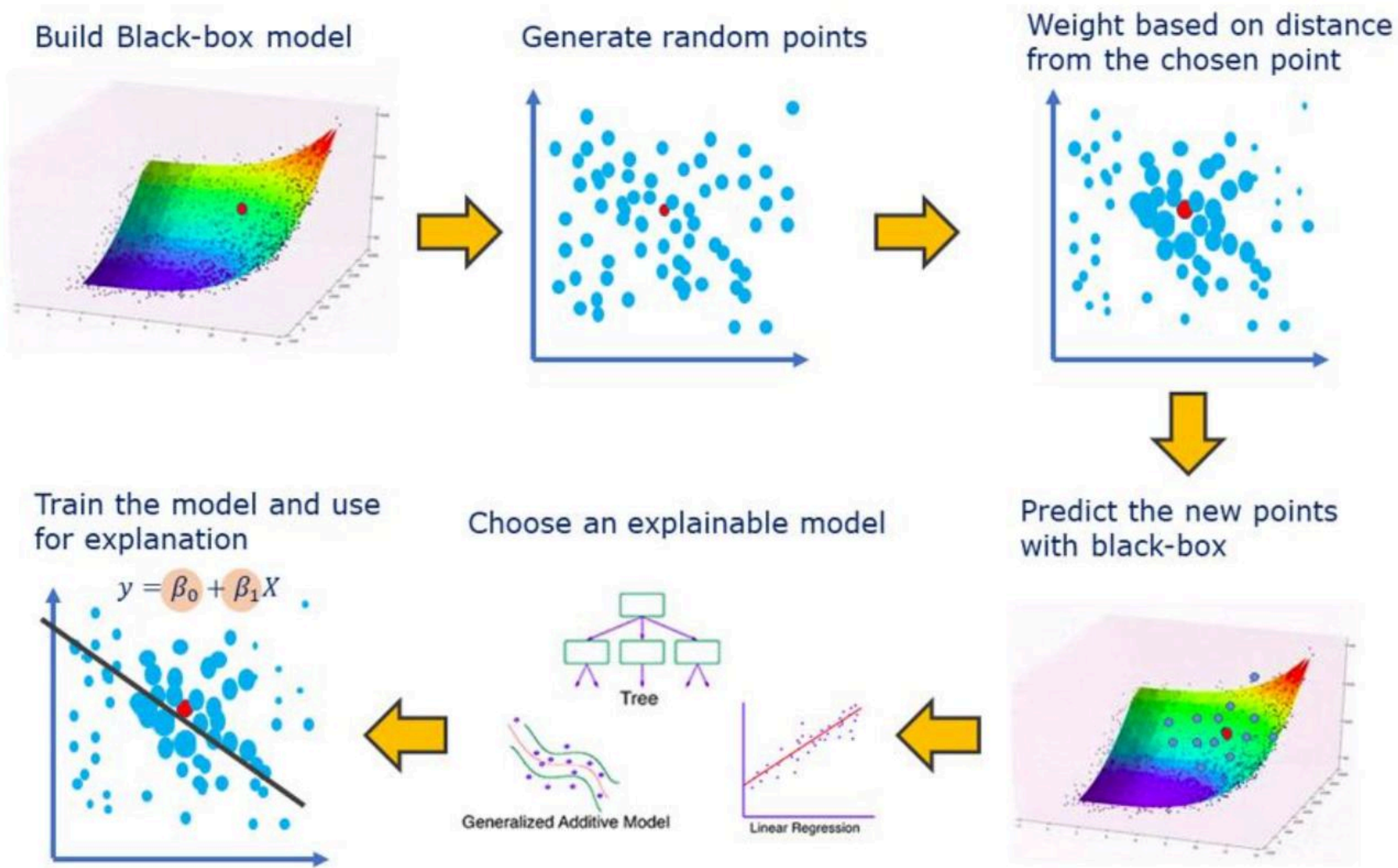
Accumulated local effects (ALE)



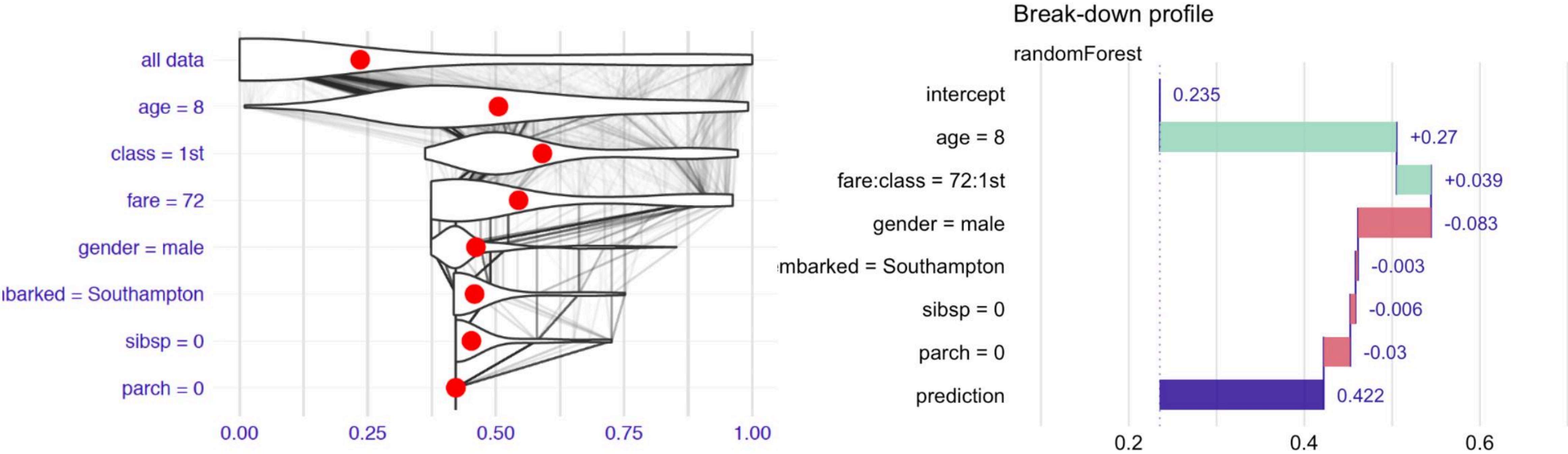
Shapley additive explanations (SHAP)



LIME



Break down plots



CH 03

DNNs and PyTorch recap

Introduction into Neural Networks and PyTorch hands-on



PyTorch

Start Locally

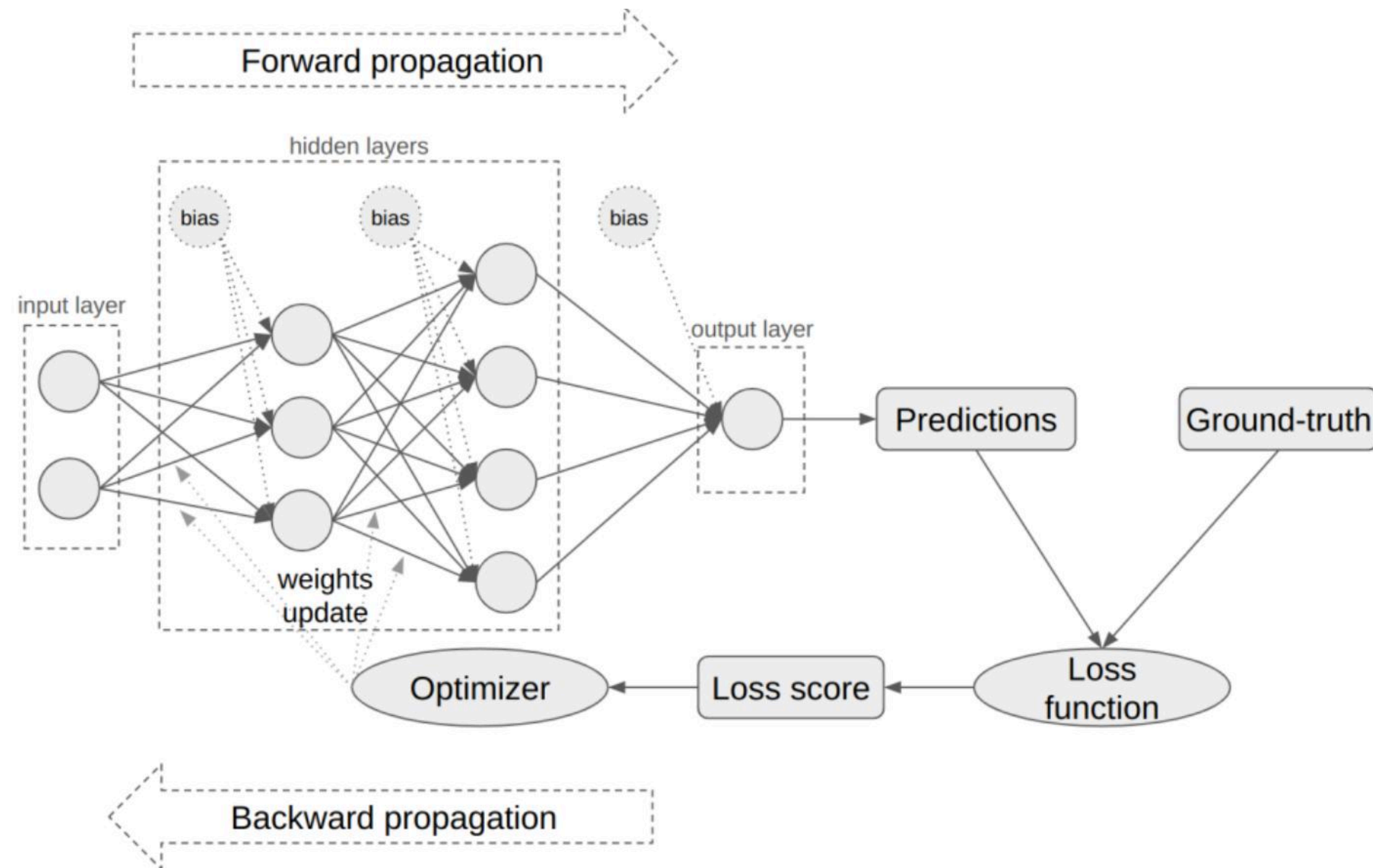
Select your preferences and run the install command. Stable represents the most currently tested and supported version of PyTorch. This should be suitable for many users. Preview is available if you want the latest, not fully tested and supported, builds that are generated nightly. Please ensure that you have **met the prerequisites below (e.g., numpy)**, depending on your package manager. You can also [install previous versions of PyTorch](#). Note that LibTorch is only available for C++.

NOTE: Latest PyTorch requires Python 3.9 or later.

PyTorch Build	Stable (2.7.0)		Preview (Nightly)		
Your OS	Linux		Mac		Windows
Package	Pip		LibTorch		Source
Language	Python			C++ / Java	
Compute Platform	CUDA 11.8	CUDA 12.6	CUDA 12.8	ROCm 6.3	Default
Run this Command:	pip3 install torch torchvision torchaudio				

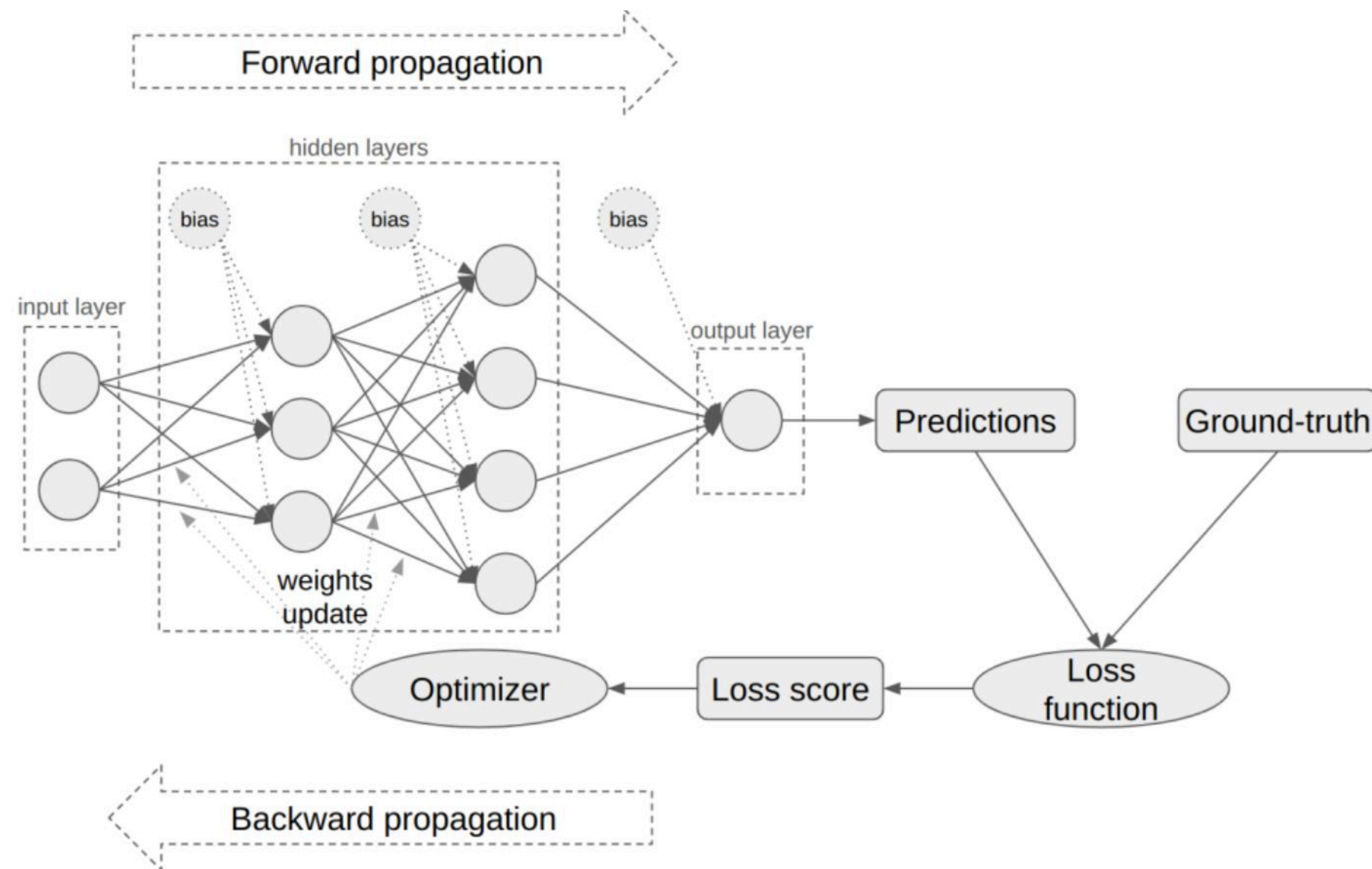
<https://pytorch.org/get-started/locally/>

Neural Network architecture



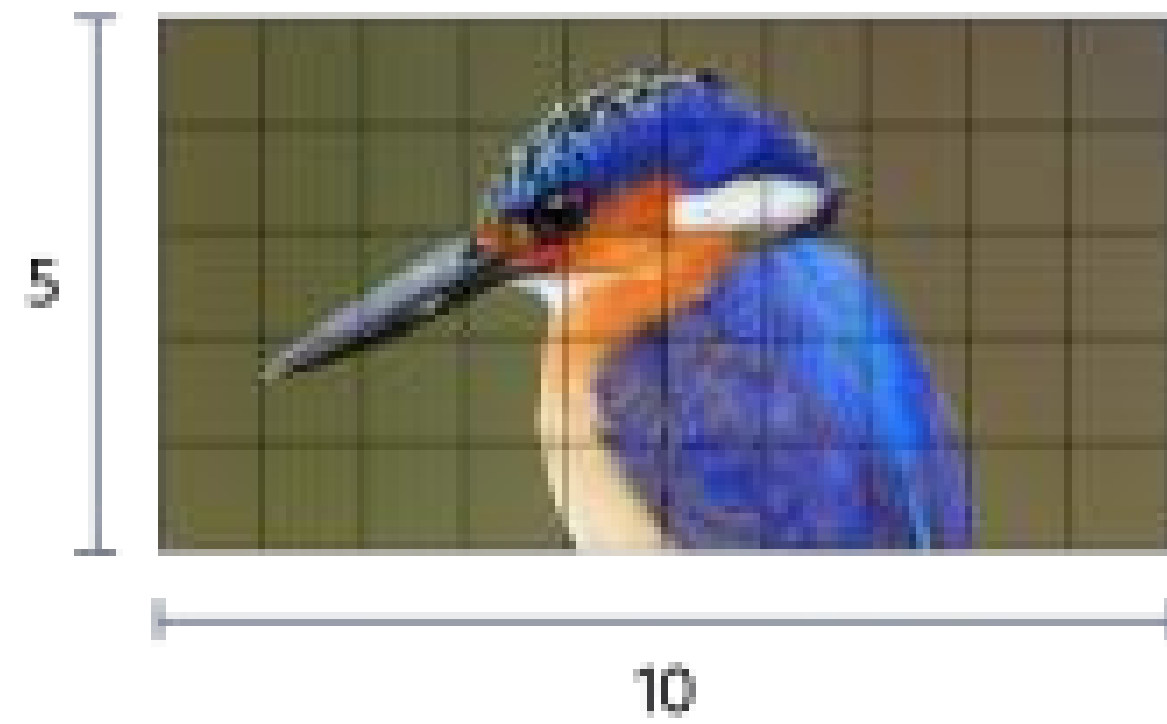
Hryniewska-Guzik, W. (2024). A multi-level perspective on the deep learning models and human-oriented explanations with applications to medical images (Doctoral dissertation). Warsaw University of Technology, Warsaw, Poland.

Neural Network architecture

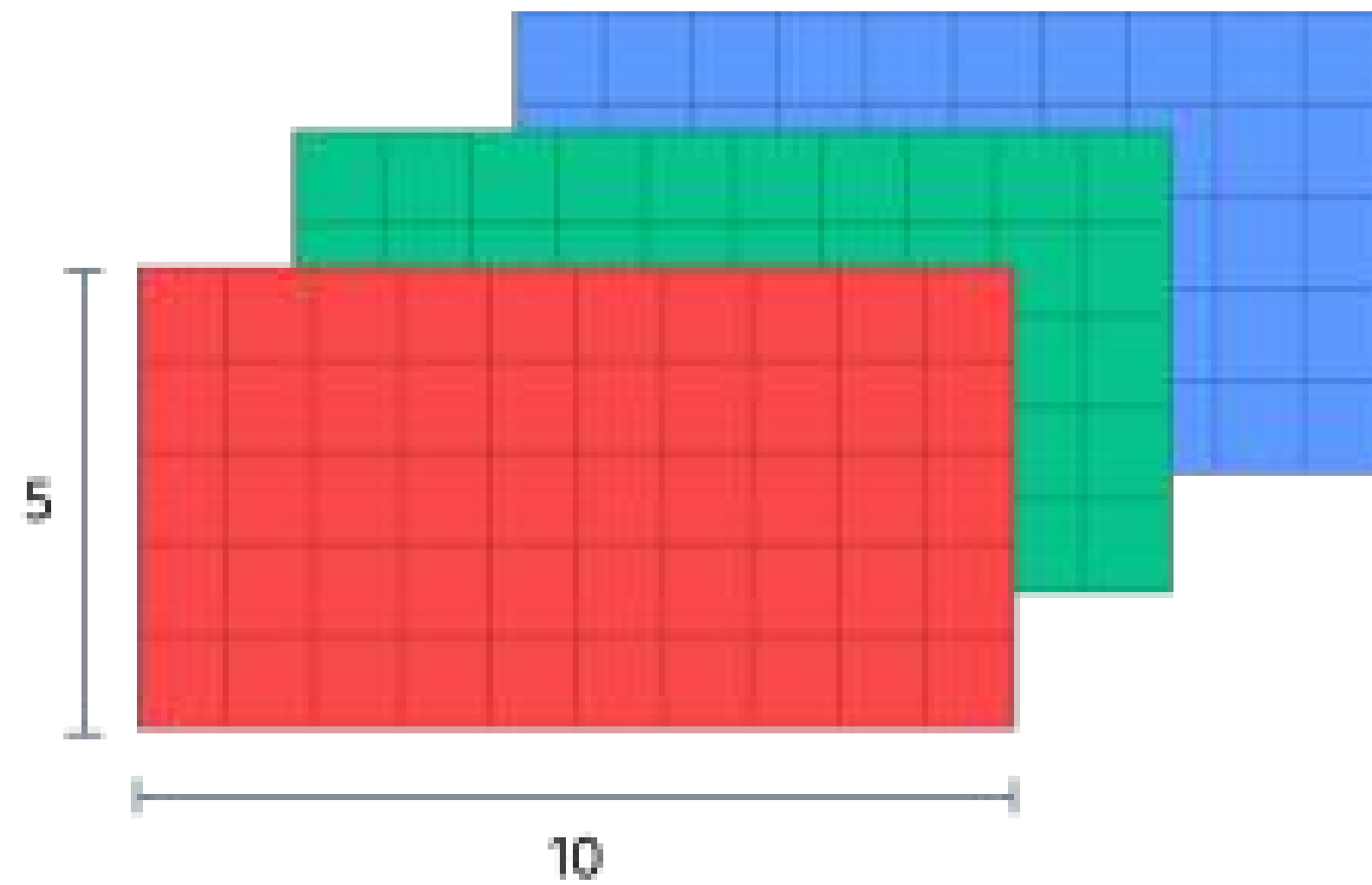


Hryniewska-Guzik, W. (2024). A multi-level perspective on the deep learning models and human-oriented explanations with applications to medical images (Doctoral dissertation). Warsaw University of Technology, Warsaw, Poland.

What are images anyway?



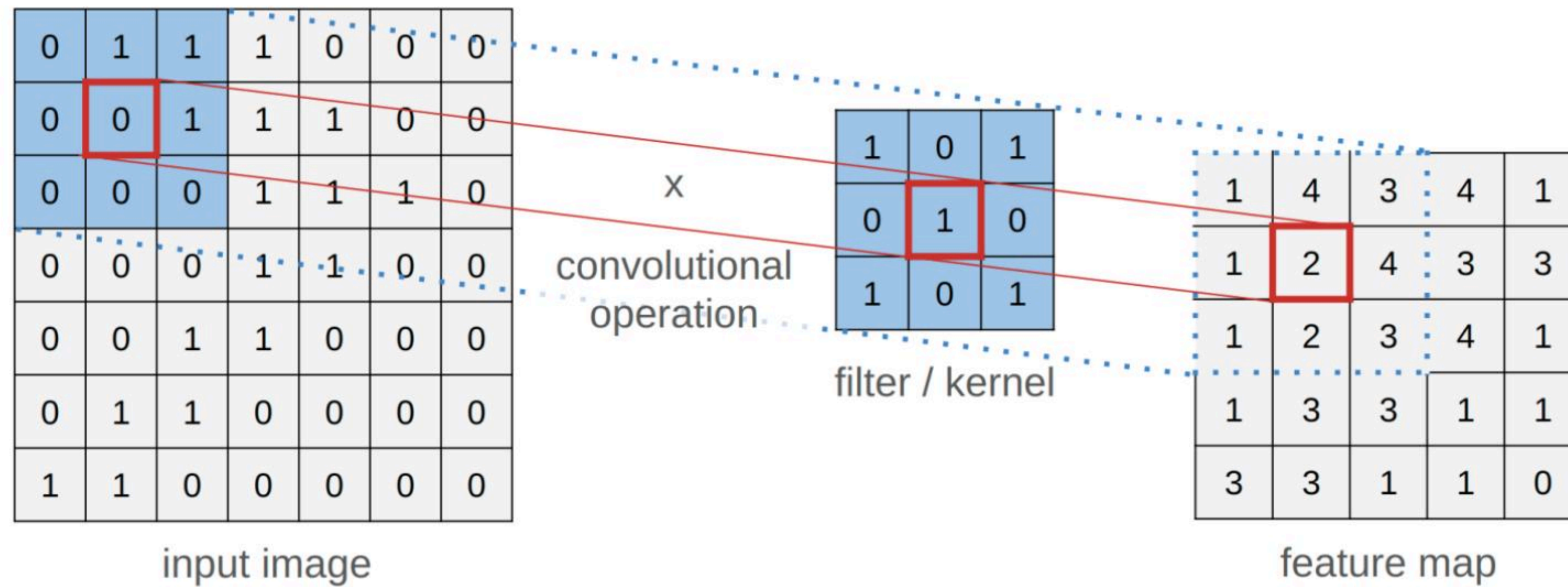
Original Color Image



3D RGB Tensor

A **tensor** is a generalized mathematical object that can represent data in one or more dimensions. Think of a tensor as a multidimensional array.

Convolution Operation



Hryniewska-Guzik, W. (2024). A multi-level perspective on the deep learning models and human-oriented explanations with applications to medical images (Doctoral dissertation). Warsaw University of Technology, Warsaw, Poland.

CH 04

Explainers for image data



Main types of explainers

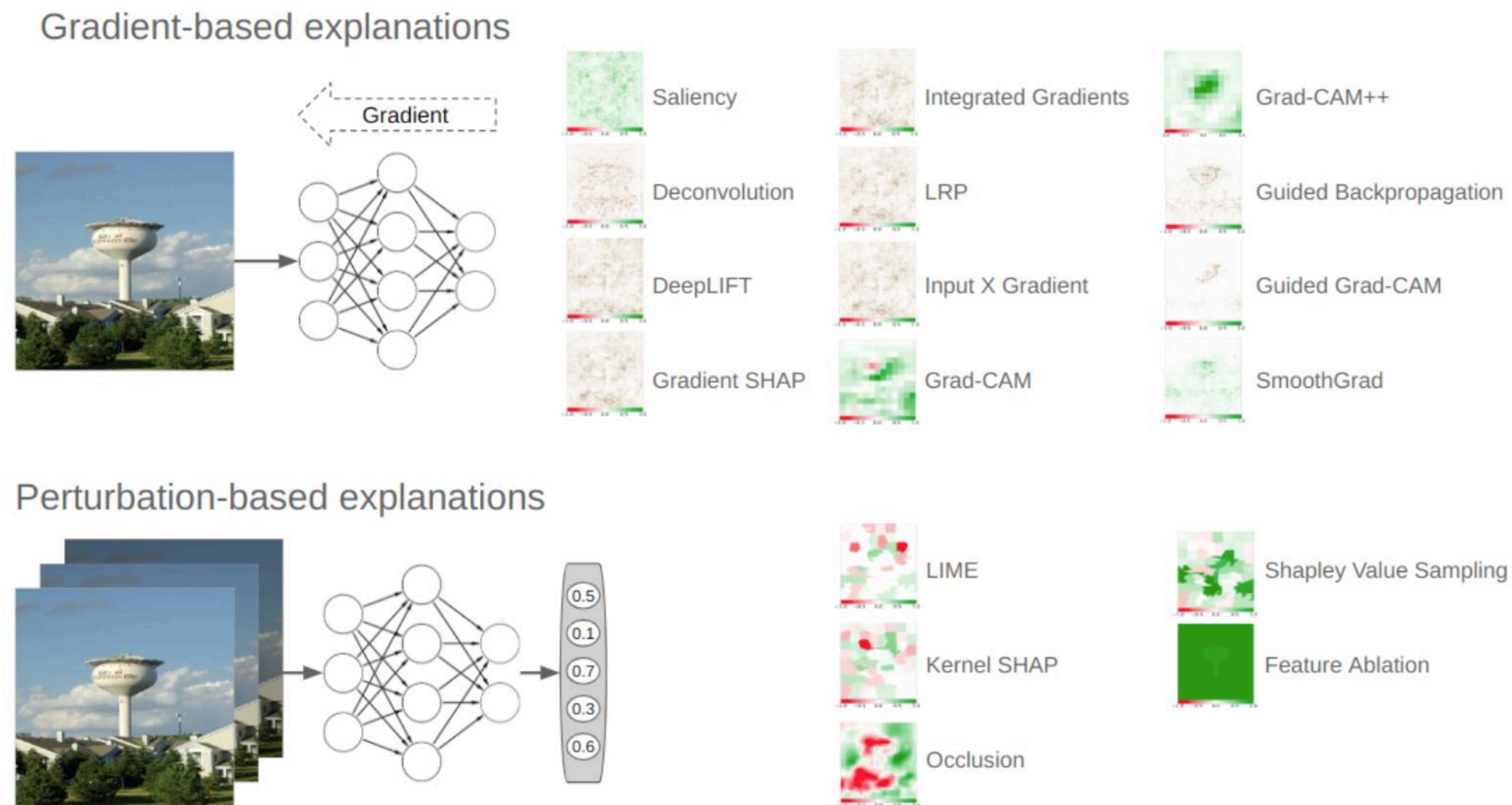
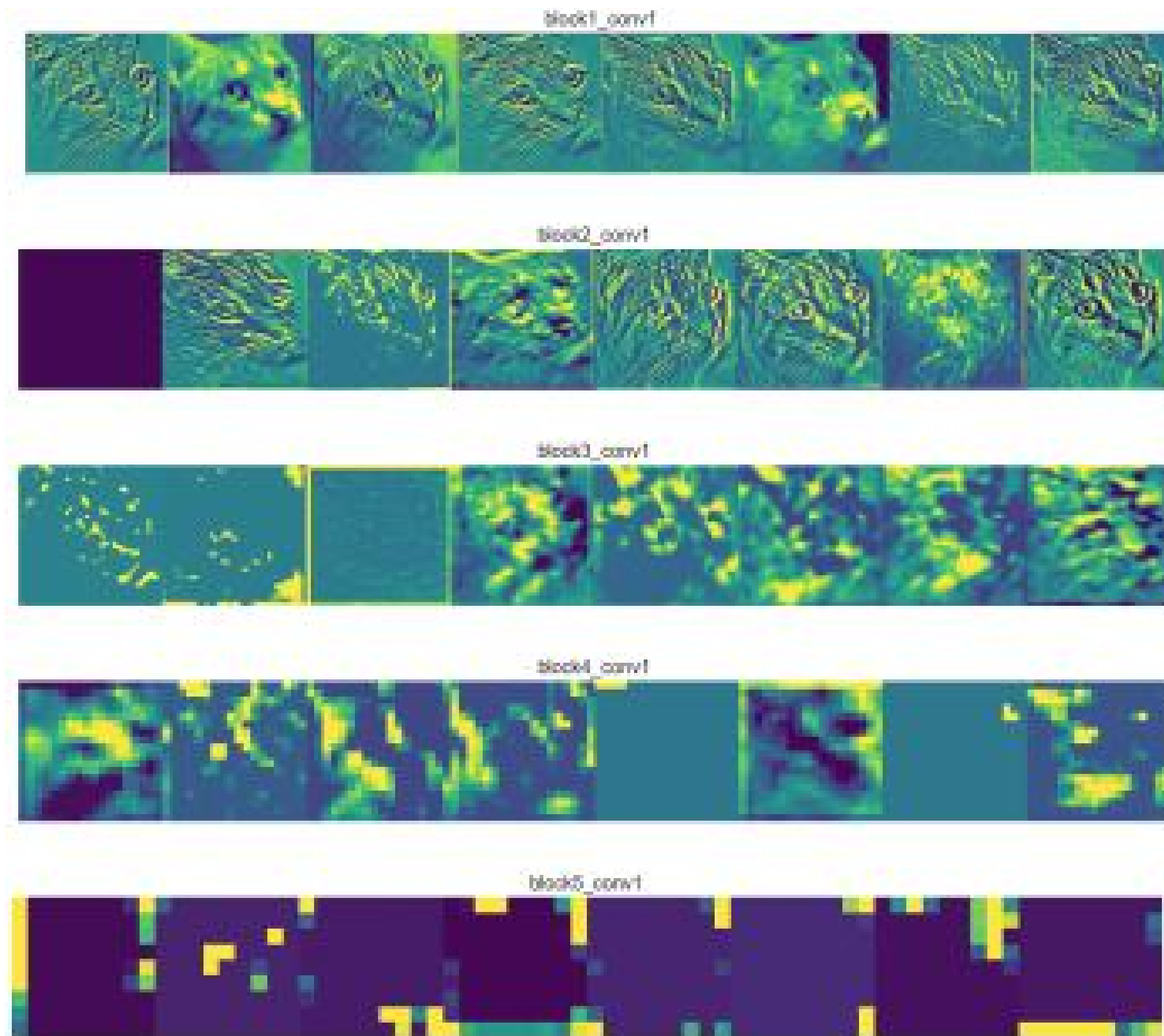


Figure 2.14: Schematic overview of gradient-based and perturbation-based methods. Heatmaps are generated using various explainability methods for SqueezeNet model pre-trained on the ImageNet dataset.

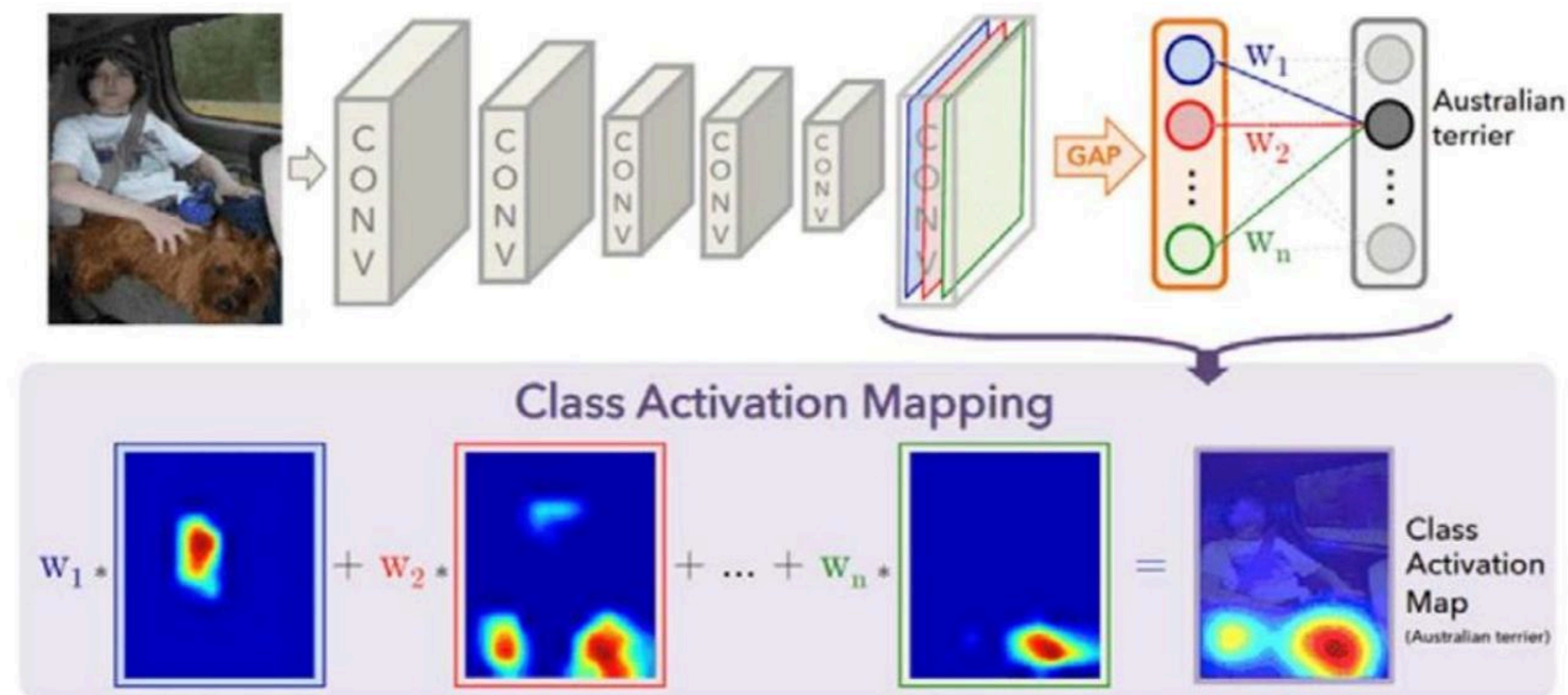
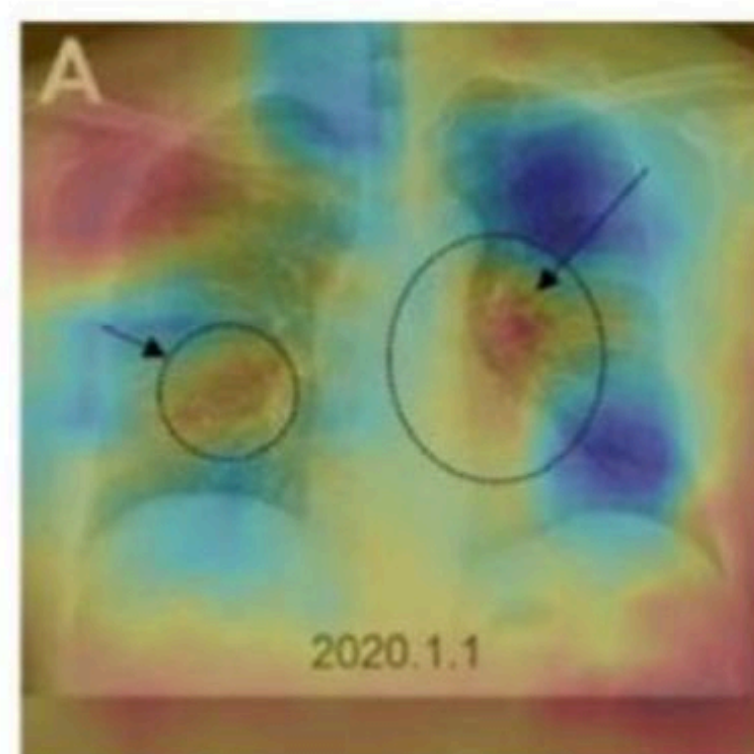
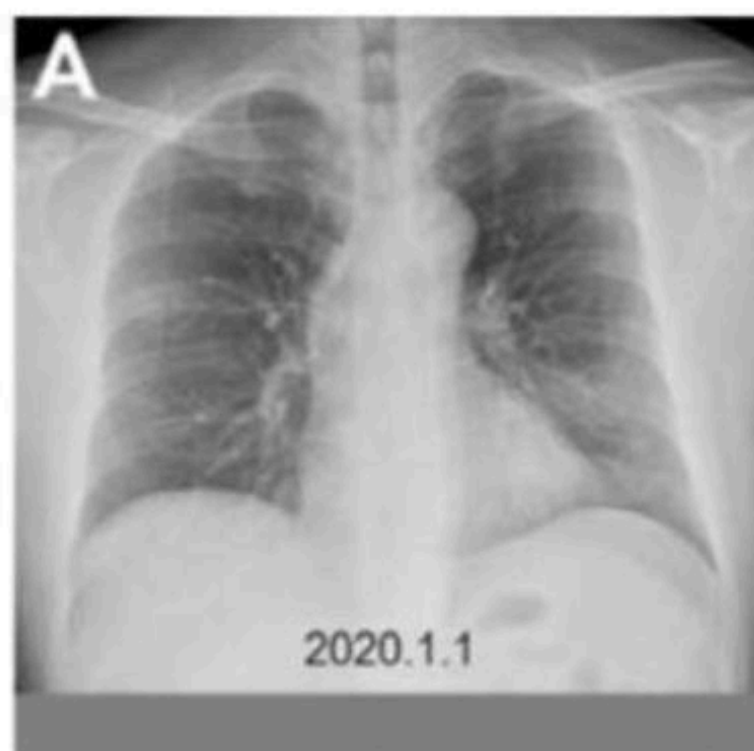
Feature maps



Convolution Operation

In CNNs, convolutional layers are used to detect and extract various features from the input data. These layers slide small filters (kernels) across the input image or the previous layer's feature map. At each location, the filter computes a dot product between itself and the corresponding region of the input. This operation results in a single value in the output, and these values collectively form the feature map.

Class Activation Mapping (CAM)



Zhou, B., Khosla, A., Lapedriza, A., Oliva, A., & Torralba, A. (2016). Learning Deep Features for Discriminative Localisation," 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR) , 2921-2929, [https:// doi.org/10.1109/CVPR.2016.319](https://doi.org/10.1109/CVPR.2016.319)

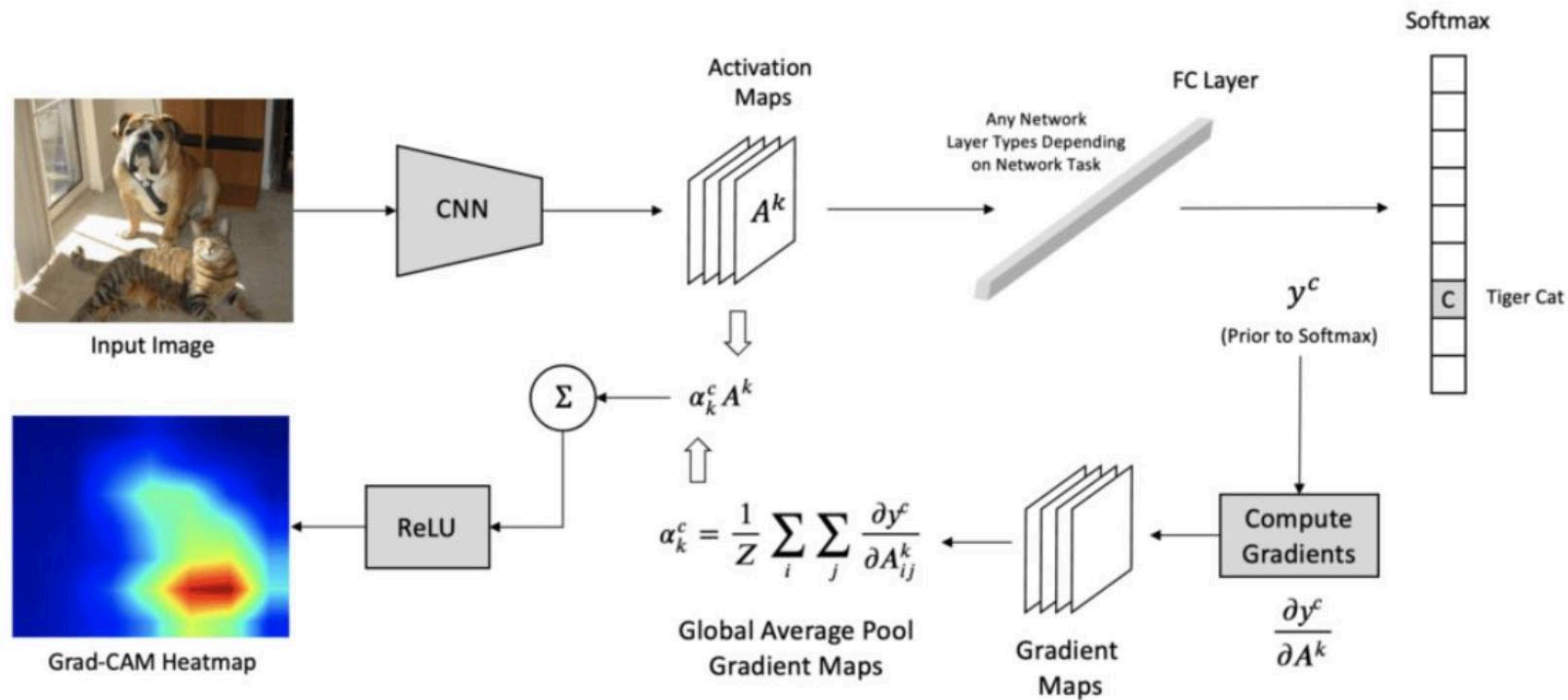
Grad-CAM

Grad-CAM (Gradient-weighted Class Activation Mapping) is an extension based on CAM, which uses the gradients for the target class that derives in the final convolutional layer. Unlike CAM, this method does not require any retraining and is broadly applicable to any architecture based on convolutional neural networks (CNN).

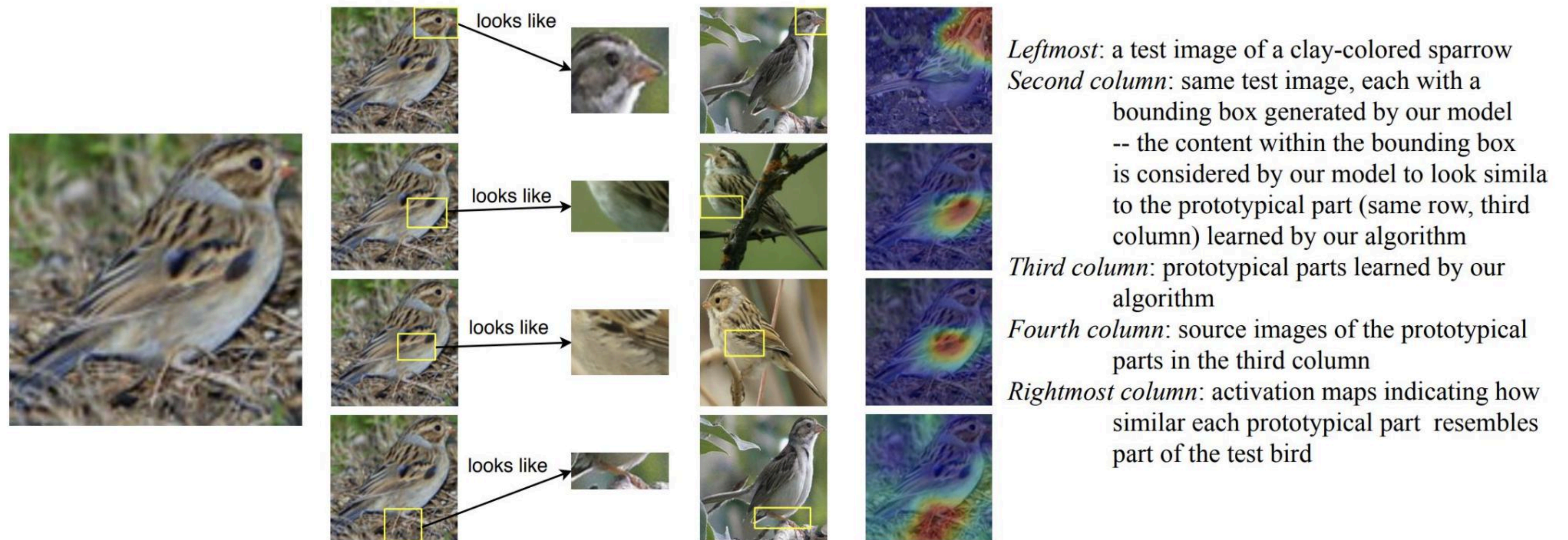
First, the class score gradient is calculated for the activation maps in the last convolutional layer. The gradients are returned after averaging them over the size of the activation map, and then the importance weights are calculated.



Grad-CAM

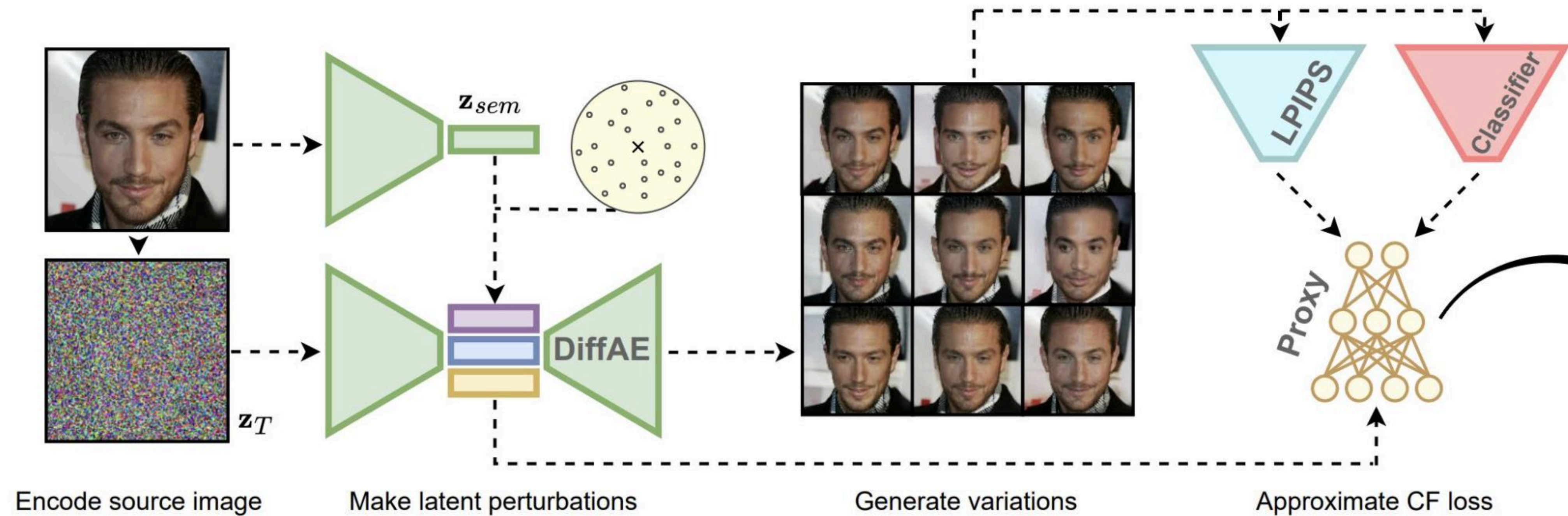


Prototypical explanations



Chen, C., Li, O., Tao, D., Barnett, A., Rudin, C., & Su, J. K. (2019). This looks like that: deep learning for interpretable image recognition. *Advances in neural information processing systems*, 32.

Counterfactual explanations



Sobieski, Bartlomiej, and Przemyslaw Biecek. "Global counterfactual directions." European Conference on Computer Vision. Cham: Springer Nature Switzerland, 2024.